

Revision Cover Letter

Date: April 2026

To: Editorial Office, *Array* (Elsevier)

Re: Major revision of “CLOP-DiT: Structured-Metadata-Conditioned Single-Cell Latent Generation via Contrastive Language–Omics Pretraining and Diffusion Transformers”

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Dear Editors,

Please find attached our revised manuscript, a marked copy, and a point-by-point response to the reviewers. The reviews were detailed and in several places uncomfortable; the revision is stronger for having to answer them.

Summary of Revision Work

The changes fall into four groups.

1. Tightened the scope and claims

- The abstract and discussion now state what CLOP-DiT is: a controllable generator of structured-metadata-conditioned cell latents, not a substitute for real single-cell data.
- Wording that overreached has been retired — claims about template optimality, “understanding” of biological semantics, and the framing of rare-cell failure modes.
- Stage-specific CLOP alignment metrics are now separated explicitly from end-to-end generation quality; the two should not have been read interchangeably in the original.

2. New analyses that address reviewer concerns directly

- Latent-space KNN error taxonomy — 49.2% of mis-typings stay inside the same biological lineage family rather than crossing into unrelated cell types.
- Organism-stratified evaluation — human-only and mouse-only types are statistically indistinguishable on all five headline metrics ($p > 0.14$).
- The phrase “biologically difficult” has been replaced with a quantitative diagnosis: within-type heterogeneity predicts fidelity better than training-set abundance ($\rho = -0.77$ vs. $+0.57$).
- Residual plots and variance-ratio summaries are now interpreted explicitly in the main text, not left to the figure alone.
- The rare-cell failure is localised: latent under-dispersion ($0.50\times$ real within-type variance) rather than decoder collapse ($1.07\times$ expression-space variance).

3. New experiments where evidence was thin

- Gaussian variance-recovery baselines that bracket CLOP-DiT’s cross-dataset r_{var} between a pooled-Gaussian floor and a type-aware oracle.
- A formal ZCA whitening ablation — three matched training runs — showing that ZCA contributes a modest ($+4.2$ pp positive-pair cosine) rather than a dramatic effect.
- An augmentation-strategy sweep plus a forced-scarcity control: no strategy helps when the baseline F1 is already at the 0.92 ceiling, but all strategies rescue F1 under genuine scarcity. The original “augmentation is ineffective” claim was too broad.

- A bridge experiment from CLOP-stage ablations to end-to-end generation. Stage-1 rankings can partially reverse (e.g., +10.2 pp Stage-1 accuracy $\rightarrow$ -45% coverage); they are not safe proxies for final quality.

4. Extensions beyond the original in-distribution scope

- Matched-dimensionality encoder comparison — the severe within-type compression is specific to scGPT (within-L2 ratio 0.14–0.07), not generic to 512-d embeddings (PCA 0.90).
- Zero-shot strict-OOD evaluation on kidney, cerebellum, and fetal gonadal tissue. The generator partially transfers along familiar structural programs (adrenal cortex 0.875, kidney epithelial 0.460) but fails on structurally distinct cerebellar neurons (≈ 0.00). This is a mechanistic limit, stated as such.
- Figures 1 and 2 have been regenerated for readability; crowded visual elements in the results figures have been rebalanced.

Key Manuscript Changes

- **Abstract** (Comment 2.1): Added research-background framing and challenge statement for label-conditioned generation.
- **Results — new analyses** (Section 3.4, Per-Type Generation Quality): KNN error taxonomy over a 10-family biologically-grounded lineage map (new appendix confusion-matrix figure), organism-stratified evaluation across human-only and mouse-only types (new appendix 6-panel boxplot), and abundance–fidelity correlation statistics added to the main text.
- **Results — variance and gene-level fidelity** (Section 3.5, Gene-Level Fidelity): Added Gaussian variance-recovery baselines and an explicit DivR vs. r_{var} distinction paragraph (supported by the latent-variance and gene-gene correlation figure embedded in the Discussion).
- **Results — rare-cell augmentation** (Section 3.11, Rare Cell Augmentation): Expanded rare-cell failure-mechanism analysis and added an augmentation-strategy sweep with a forced-scarcity control (new appendix F1-trajectory figure).
- **Ablation section** (Section 3.10, CLOP Ablation Study): Added formal ZCA whitening ablation (new main-text Table 6, *tab:zca-ablation*) and a CLOP $\rightarrow$ full-pipeline bridge experiment with explicit partial-reversal caveat reported inline in the existing Table 5 (*tab:ablation*) context.
- **Conditioning field ablation** (Section 3.12, Conditioning Field Ablation and Swap-Label Permutation Test): Expanded Table 7 (*tab:cond-ablation*) to include field ablations and swap-label permutation evidence cited for the Gaussian benchmark reply.
- **Discussion**: New unified mechanistic narrative (latent-stage under-dispersion is the bottleneck), encoder bottleneck diagnosis, Gaussian benchmark framing clarification, and an updated strict-OOD paragraph reflecting the new zero-shot evaluation.
- **New Results Section 3.13 Strict-OOD Generalization (Zero-Shot)**: Per-tissue and per-cell-type zero-shot evaluation on kidney, cerebellum, and fetal gonadal tissue (new main-text Table 8, *tab:strict-ood*, and Figure 9, *fig:strict-ood*).
- **Figures 1–2** (Comment 2.4): Regenerated with improved readability—larger fonts, wider component spacing, increased background contrast, and consistent line weights.
- **New experimental artefacts introduced in the revision**: three main-text additions (Table 6 ZCA ablation *tab:zca-ablation*, Table 8 strict-OOD summary *tab:strict-ood*, Figure 9 strict-OOD per-cell-type bar chart *fig:strict-ood*) and three new Supplementary Figures in Appendix A (KNN family heatmap *fig:knn-family-heatmap*, organism-stratified boxplot *fig:organism-stratified*, forced-scarcity F1 curves *fig:forced-scarcity*).
- **Language**: Replaced vague terms (“biologically difficult,” “understands biological semantics,” “optimal five-field template”) with precise operational definitions throughout.

Accompanying Documents

1. **Revised manuscript**
2. **Marked manuscript**
3. **Point-by-point response letter** — detailed responses to all 16 reviewer items (11 from Reviewer #2, one additional concern on the Gaussian benchmark, and 4 from Reviewer #3) with original reviewer text quoted, quantitative evidence, and manuscript cross-references
4. **This revision cover letter**

The revised submission is more precise about what CLOP-DiT does, what it does not yet do, and where the next scientific effort should go. We are grateful for the reviewers' time.

Sincerely,

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on behalf of Zeyu Fu, Yiyao Liu, Junping Wang, and Song Wang